Actual-Source Arithmetic and Undecidability Boundaries for Canonical Comparison

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Abstract

We relate universal comparison of canonical Wilken–Weiermann assignments to existential arithmetic of actual typed sources. A constant-size addition gadget, a superincreasing sorting construction and an asymmetric collision detector compile arbitrary finite Boolean formulas into one unconditional ground comparison, preserving shared witnesses. Absolute observers also handle functional-output sources. We construct original-budget source certificates through an entire grammar with arbitrary nested ground binders, and eliminate functional unit observations by finite systems of actual source equations. Finally we prove two undecidability boundaries: functional free variables at the unchanged standard control, and ground-only inputs after an additive-one perturbation of that control. The exact standard-control all-ground comparison problem is not classified by these results.

Contents

1 Source observations are an arithmetic language **2**

2 Literal arithmetic and collision gadgets **2**

3 A full Boolean compiler **3**

4 Functional outputs and absolute observation **4**

5 Eliminating functional unit observations **5**

6 A complete source grammar with ground binders **6**

7 From a certificate to the whole original head graph **8**

8 Functional inputs at the unchanged control **9**

9 An additive-one change of control **12**

10 The remaining exact-control boundary **14**

1 Source observations are an arithmetic language

The assignment is the literal zero-history assignment of original Gödel T terms, with the recursor convention of Wilken–Weiermann [1]. The foundational companion [4] gives its full clauses, admissible substitutions, exact closed evaluator and generated-source forward-translation theorem. Ordinal addition and multiplication in positive fields are natural; scalar fields are ordinary positive natural numbers. Initially the control is fixed to

$$F_0(t) = 2^t, \quad F_{j+1}(t) = F_j^{t+1}(t), \quad \Phi(t) = F_5(t + 100).$$

A source equation below means equality of canonical scalar values, not beta equivalence of programs. All bound names introduced by a construction are fresh, and all substitutions are capture avoiding.

Every free ground coordinate ranges over $\mathbb{N}_{>0}$. To work with usual natural arithmetic we encode $n \in \mathbb{N}$ by the raw source coordinate $n + 1$. The literal source constant 0 has assigned value one, so represents arithmetic zero. Repeated free names always have the same valuation.

Definition 1.1 (Actual-source arithmetic)**.** Let $\mathcal{E}_{\mathrm{src}}$ be existential arithmetic over $\mathbb{N}$ with constants, addition, equality, order and finite Boolean combinations. An extra function may be given by a supplied closed original unary source $H : o \to o$:

$$F_H(n) = [\![HX]\!]_0 - 1, \qquad x_0 = n + 1.$$

We also allow an observation leaf $\mathrm{Source}(A) = [\![A]\!]_0 - 1$ for any supplied ground-output original source with ground free names, using the same natural/raw encoding for those names.

The source syntax is part of the finite input. One does not adjoin every ordinal-indexed collapse merely because its notation is computable. The functions in this language are total, but their simultaneous existential theory is not known to be decidable.

2 Literal arithmetic and collision gadgets

Write $g(n) = \psi(\omega + n) = \Phi(n + 2) + 1$. From the canonical clauses, for a fresh unused ground binder define

$$e(A, B) = (\lambda u : o.A)B, \qquad [\![e(A, B)]\!]_0 = g(3[\![A]\!]_0 + [\![B]\!]_0 + 2).$$

This uses the full canonical assignment rather than evaluating the beta reduct. The fixed-tail separation lemma of [4] makes g injective and gives $g(n + 1) > 4g(n)$ for $n \geq 6$.

Lemma 2.1 (Addition and small constants)**.** *Natural addition $n_a + n_b = n_c$ is defined by the two source equations*

$$e(b, 0) = e(0, d), \qquad e(a, d) = e(c, 0),$$

with one fresh positive raw witness d. Arithmetic one has a constant-size source-equation definition. Every binary literal k has an addition DAG of size $O(\log(k + 1))$ once zero and one are available.

Proof. Injectivity gives $d = 3b - 2 \geq 1$ and $a + b = c + 1$. The latter is exactly the natural addition equation after subtracting one from each raw coordinate. The witness is unique, and any true addition supplies it.

Let $I = \lambda z : o.z$. The prime and lambda clauses give $[\![S]\!] = (1, 1)$ and $[\![I]\!] = (3, 1)$. The equation $Sv = I0$ forces raw $v = 3$. Adding $\mathrm{Add}(u, u, v)$ forces raw $u = 2$, representing natural one. Its addition witness is $d = 4$. Thus no giant closed value has to be evaluated to define one. Repeated doubling and add-one gates give the usual binary construction of k. □

A supplied graph call $F_H(n_a) = n_b$ is simply the source equation $Ha = b$. A source leaf is flattened by $A = y$. These equations retain the actual supplied source and every shared input occurring inside it.

Lemma 2.2 (Superincreasing pair coding)**.** *Let $q(n) = F_S(n) = \Phi(n + 4)$. Then*

$$q(a) + q(b) = q(c) + q(d) \quad \Longleftrightarrow \quad \{a, b\} = \{c, d\}$$

as multisets. Consequently sorting two naturals has a primitive-positive definition using addition and the actual S graph.

Proof. The control bound gives $q(n + 1) > 2q(n)$. Cancel equal indices across the two sides. If a largest remaining index $M \geq 1$ occurs, the other side has at most two values at most $q(M - 1)$, too small to equal $q(M)$. If only index zero remains on one side, its positive contribution also cannot vanish. This proves multiset equality; the converse is immediate. Impose

$$q(a) + q(b) = q(l) + q(h), \qquad l + \delta = h.$$

Multiset equality and nonnegative δ force $l = \min(a, b)$, $h = \max(a, b)$ and $\delta = |a - b|$. These values conversely satisfy the equations. Flattening the sums and graph calls uses only the previous lemma. □

Lemma 2.3 (Six-eraser collision detector)**.** *For arbitrary actual ground sources P, Q, put*

$$L = e(e(P, P), e(Q, Q)), \qquad R = e(e(P, Q), e(Q, P)).$$

At every joint valuation $L_0 \geq R_0$, with equality exactly when $P_0 = Q_0$. Otherwise $L_0 - R_0 \geq 2$.

Proof. For raw $p = P_0, q = Q_0$ set

$$A = g(4p + 2),\ B = g(4q + 2),\ C = g(3p + q + 2),\ D = g(p + 3q + 2).$$

Then $L_0 = g(3A + B + 2)$ and $R_0 = g(3C + D + 2)$, with all inner arguments at least six. If $p > q$, then $A > 4C$ and $C \geq D$, giving $3A + B > 3C + D$. If $q > p$, then $B > 4D$ and $D \geq C$, giving the same strict conclusion. For $p = q$ every inner value agrees. The fourfold growth bound on the outer g also gives the integer gap at least two. Thus the unequal weights three and one do not require a symmetric source adder. □

Lemma 2.4 (Conjunction without cancellation)**.** *A finite conjunction of actual ground-source equations can be transformed into ground sources L, R with $L_0 \geq R_0$ everywhere, equality precisely when every equation holds, and otherwise gap at least two.*

Proof. Apply the collision detector to each equation. Combine all left outputs and all right outputs in identically shaped e-trees. Strict increase in both arguments preserves nonnegativity, and equality survives exactly when it survives at every leaf. One unequal leaf makes its ancestors strict; the growth bound gives the final gap. A balanced tree adds only linear DAG overhead. For an empty conjunction take $L = R = 0$. □

3 A full Boolean compiler

Theorem 3.1 (One unconditional comparison)**.** *Every existential $\mathcal{E}_{\mathrm{src}}$ sentence has a computably constructed ground-output strict comparison, and one weak comparison, true precisely when the sentence is false. All free variables of the resulting sources are ground. Shared inputs and source leaves retain their valuations. The added source DAG size is polynomial in the formula DAG, binary constants and supplied source DAGs.*

Proof. Flatten the arithmetic terms once, sharing each original DAG node. The addition and supplied-source graph equations are conjoined outside the Boolean matrix. They define total functions, so this operation preserves truth even under negation.

For each Boolean node construct a natural bit b. To test equality $a = c$, sort a, c and obtain $\delta = |a - c|$; sort $\delta, 1$ and let $v = \min(\delta, 1)$; require $b + v = 1$. Thus $b = 1$ exactly when the inputs are equal. Weak order $a \leq c$ is the equality test $\max(a, c) = c$. Strict order is the complement of the reverse weak-order bit. Negation uses $b + c = 1$, conjunction uses $b = \min(c, d)$, and disjunction uses $b = \max(c, d)$. The preceding sorting lemma defines each operation by a constant number of equations. Induction proves that every output is in $\{0, 1\}$ and has exactly its intended truth value.

Require the root bit to be one. For each valuation of the original inputs, the flattened values, sorting witnesses, differences and bits are uniquely determined, and supply a solution exactly when the matrix is true. Addition witnesses are positive by Lemma 2.1. This proves both directions without a disjunctive-normal-form expansion or an unbounded integer witness. Repeated source names have never been split into independent copies.

Apply Lemma 2.4 to the resulting conjunction. Since every existential witness is now a free ground coordinate, universal $R \prec L$ is exactly absence of a witness. For the weak version put $K = \lambda z : o.0$, with $[\![K]\!] = (5, 1)$. Literal application gives

$$[\![KR]\!]_0 \leq [\![IL]\!]_0 \quad \Longleftrightarrow \quad g(5 + R_0) \leq g(3 + L_0) \quad \Longleftrightarrow \quad L_0 - R_0 \geq 2.$$

Equality or a gap of two are the only alternatives. The weak comparison therefore expresses the same unsatisfiability assertion.

Every arithmetic or Boolean node adds constant gadget overhead, binary constants add logarithmic overhead, and the final detector/aggregation adds linear overhead. Existing source DAGs remain shared. This proves the polynomial DAG claim, not a polynomial expanded-tree or evaluation-time bound. □

Corollary 3.2 (Ground weak/strict equivalence)**.** *On all actual ground-output sources with ground free variables, universal weak and strict comparison are polynomial-DAG many-one interreducible.*

Proof. For a weak query $A \preccurlyeq B$, compile the existential counterexample $\mathrm{Source}(A) > \mathrm{Source}(B)$ into the strict version. For a strict query compile $\mathrm{Source}(A) \geq \mathrm{Source}(B)$ into the weak version. No recognition of a positive arithmetic fragment is needed, because the compiler accepts arbitrary supplied ground-source leaves. □

4 Functional outputs and absolute observation

The generated-source invariant in [4] states that positive coordinates are independent of ground inputs. Capture-free substitution of a closed ground term also commutes with a canonical assignment: this is structural induction, using the distinct-name partial substitution lemma of [1, Lemma 2.20] at abstraction. No reduction history is changed.

Lemma 4.1 (Absolute observer)**.** *Let $T : \rho$ have only ground free names $\mathbf{z}$. Its entire raw scalar graph $v = [\![T(\mathbf{z})]\!]_0$ is existentially definable by actual ground-source observations, addition and constants, with all shared names preserved.*

Proof. Close the ground names to literal zero, obtaining T^*, and evaluate the closed scalar $C = [\![T^*]\!]_0$. Put $m = \mathrm{lv}(\rho)$ and $p = 2(m + 1) + 1$. For a fresh unused ground u, the full vector

size multiplier gives

$$[\![\lambda u : o.T]\!]_0 = p[\![T]\!]_0 + p - 1.$$

The positive profiles agree for T and T^*. With fresh positive ground inputs x, y, form the actual ground sources

$$G(\mathbf{z}, x) = (\lambda Y : \rho.0)((\lambda u : o.T)x), \qquad G^*(y) = (\lambda Y : \rho.0)((\lambda u : o.T^*)y).$$

The inner applications and outer erasers use matching fixed heads on the two sides. Twice cancelling their injective fixed-tail collapses yields

$$G_0 = G_0^* \iff p[\![T]\!]_0 + x_0 = pC + y_0.$$

Add $pv + x_0 = pC + y_0$. These two equations force $v = [\![T]\!]_0$; conversely raw witnesses $x_0 = pC + 1$, $y_0 = pv + 1$ work. All raw values are encoded by naturals plus one. The anchor C is effectively computed by the closed evaluator; that cost is charged, with no feasible-size claim. □

Theorem 4.2 (Exact decision equivalence)**.** *Universal weak/strict canonical comparison on the full all-ground-free-variable domain, its ground-output restriction, and unsatisfiability in $\mathcal{E}_{\mathrm{src}}$ are computably equivalent. Satisfiability uses answer complementation.*

Proof. For a general-output comparison first decide its closed positive-coordinate tests. If they pass, a countervaluation is an existential inequality between the scalar observations defined by Lemma 4.1. The Boolean compiler handles that sentence. Conversely it reduces arbitrary source arithmetic unsatisfiability to ground-output comparison. The ground-output domain is a special case of the general domain. The closed-anchor evaluations in the general-output direction justify computability, not a uniform polynomial bound. □

The equivalence isolates a real open problem. The formula compiler is complete; it does not decide whether its witnesses exist. A faster closed evaluator, a graph for one more source head or an ordinal termination bound does not supply that missing negative-instance decision procedure.

5 Eliminating functional unit observations

A unit observation substitutes the number one in every coordinate of a free functional variable. It is an observation of the assignment, not a claim that a closed term has the all-one vector. We first show how to eliminate such observations without requiring that inhabitant.

Lemma 5.1 (One-name scalar elimination)**.** *Let $A : \tau$ be an actual source, $h = [\![A]\!]$, $m = \mathrm{lv}(\tau)$, and let $X : \sigma$ be one of its free functional names. Put*

$$n = \mathrm{lv}(\sigma) + 1, \quad d = 2^n \sum_{i=0}^{m} \mathrm{sz}^X(h_i), \quad P = (\partial_0^X h_0)_0, \quad R = h_0\{X := \mathbf{1}\}, \quad L = \lambda X.A.$$

The graph of the desired scalar R has a finite positive-coefficient linear definition using scalar observations of actual sources without free X.

Proof. The original abstraction clause gives $L_0 = dP + R$. If the whole h_0 is X-free, the stopping rule gives $P = R + 1$, hence $L_0 = (d + 1)R + d$. The positive coefficient uniquely determines R.

Suppose h_0 is active. Introduce a fresh unused ground u and put $V = \lambda u : o.A$, $k = 2(m+1)$, $p = k+1$, $h' = [\![V]\!]$, $d' = 2^n \sum_j \mathrm{sz}^X(h'_j)$ and $L' = \lambda X.V$. The literal scalar is $h'_0 = k(h_0+1)+h_0$, where the multiplication means the original finite sum of copies, before any collecting. Its partial is $pP+4k$: a copy of h_0+1 contributes $P+3$, and the k sum nodes joining the copies and retained h_0 contribute k more. The retained unit value is $pR+k$. The scalar size is $p\,\mathrm{sz}^X(h_0)+3k > \mathrm{sz}^X(h_0)$.

No positive-coordinate size decreases. For $m \geq 1$ the new first field has two copies of h_1 and finite constants, and higher fields have two copies of their old field. An entire X-free field has size one on both sides; an active field has at least its original size. For $m = 0$ the new positive field is one. Therefore $d' > d$. With $c = 4kd' + k$ we obtain

$$L'_0 = pd'P + pR + c, \qquad pd'L_0 + dc = dL'_0 + p(d'-d)R.$$

The coefficient of R is positive. These two sources have no free X, and every other functional name can still receive its common unit observation. The intended R satisfies the equation and is its unique solution once the other observations are fixed. All coefficients are finite syntactic integers. □

Theorem 5.2 (Finite unit-observation elimination)**.** *For an actual source A and a finite set of all its free functional names, the absolute scalar after their common unit observation is existentially definable in $\mathcal{E}_{\mathrm{src}}$, retaining all original shared ground inputs.*

Proof. Apply Lemma 5.1 to one name, then recursively to the actual sources L, L' on the remaining set. Each branch removes a name; the new u is ground. Induction on the number of functional names terminates. Leaves have only ground free names. Lemma 4.1 supplies their absolute scalar graphs, including leaves of functional output type.

For each fixed ground valuation the intended values supply positive witnesses. Reverse induction through the positive-coefficient equations proves uniqueness of every internal scalar. Encoding raw values as naturals plus one merely changes fixed additive constants. The Boolean compiler consumes the resulting finite system. No repeated original name has acquired an independent value. □

6 A complete source grammar with ground binders

Fix a tracked $F : \sigma_k = o^k \to o$, $k \geq 1$. Contexts of level at most one are generated by F, ground variables, supplied actual F-free blocks of result level at most one, ground-argument application, and abstraction $\lambda z : o.U$. Nesting and functional-result ground lambdas are unrestricted. An opaque block may have arbitrary internal types, lambdas and recursors, and may capture surrounding ground names. Other free functional names receive a single common unit observation.

Write $q(U) = [\![U]\!]_0$ and $h(U) = [\![U]\!]_1$ when U is functional. A scalar collapse in an active scalar branch is annotated by its actual operator source. This annotation is retained under scalar copying, ground unit substitution and ground partials. It is not an assertion that every scalar subexpression is itself a source.

Lemma 6.1 (Affine heads and their finite surplus)**.** *Every functional context has an additive first-field tree with leaves F_1 and whole F-free positive expressions. At a selected collapse with head h, its tail is F-active whenever h is active. At common units there is a finite $\delta(h) \geq 0$ such that*

$$(\partial_1^F h)_1 = h^* + \delta(h).$$

The surplus is zero only at the bare leaf F_1; a nonfinite h^* has $\delta(h) \geq 1$. The output-two head partial is finite.

Proof. Ground application retains the operator's positive fields. A ground lambda with ground body has first field $(\partial_0^z q)_1$; the whole-free and variable cases are one, a sum adds its two outputs and one, and an active collapse contributes its ground-free positive head plus one and the tail partial. For functional body the resulting first field is $(\partial_0^z q)_1 + 2h + 1$. These clauses prove additive closure by source induction, and show that occurrence of F in an operator head implies occurrence in its scalar. In a collapse its scalar tail includes that operator scalar. Ground partials and ground units preserve this support.

On the additive head tree a whole-free stop has surplus one, a tracked F_1 leaf has surplus zero, and an active sum adds its child surpluses and one. Only the bare tracked leaf has surplus zero, and its unit value is finite. Output two is one at either leaf and adds child values and one at an active sum, proving its finiteness. □

Lemma 6.2 (Actual unary carriers)**.** *Each selected nonfinite head has an actual unary carrier with that exact unit first field. Its ground captures can be closed to zero without changing the field, and its actual scalar is retained.*

Proof. If its annotated operator is $U : o^r \to o$, replace the free tracked name throughout U by one fresh same-type $J : \sigma_k$ and apply $r - 1$ literal zero arguments. This yields $V_U : o \to o$. Each ground application retains the first field. The same common unit observation on J reproduces the original head; this is global same-type renaming, not a change of the result type of one head occurrence. Open captured ground names using their fresh identities and close them to zero. The generated-source invariant proves that their values cannot affect the positive field. No scalar is reset to one. □

Lemma 6.3 (Multiset accounting and ground transport)**.** *For every annotated scalar q there are an actual-carrier multiset $\mathcal{M}(q)$ and a finite $c(q) \geq 0$ with*

$$(\partial_0^F q)_1 = \sum_{V \in \mathcal{M}(q)} V_1 + c(q), \qquad d(q) = c(q) - 3|\mathcal{M}(q)|.$$

On the generated active sources, $d \geq 1$; it is at least four except possibly at the bare functional source F. Ground units preserve the multiset and d, and a ground scalar partial preserves the multiset and does not decrease d.

Proof. Define the accounting literally. At an F-free scalar or F_0, use the empty multiset and $d = 1$. At an active sum use multiset union and $d = d_1 + d_2 + 1$. At an active collapse $\psi(\omega \otimes h + t)$, the tail is active by Lemma 6.1. For finite h^* retain its tail multiset and add $h^* + \delta(h)$ to d. For nonfinite h^* append V_U and add $\delta(h) - 3$. The canonical positive partial proves each identity; subtraction here is only in the finite accounting integer.

A ground substitution changes no positive head and retains active nodes, so its accounting is identical. For a ground partial $D_z q$, an F-free piece remains a free stop. On an active z-free q the new $q + 1$ adds two to d. An active sum is transformed into $(D_z q_1 + D_z q_2) + 1$, again adding two beyond the child contributions. At an active collapse the ground-free head and its carrier decision do not change, and induction handles the tail. This proves the transport statements for every nested ground binder.

It remains to check the lower reserve on sources. The base F has one. An active ground application has scalar $\psi(\omega \otimes h + q_U + q_V)$. If h^* is finite, its reserve is $d(q_U) + d(q_V) + 1 + h^* + \delta(h) \geq d(q_U) + d(q_V) + 2$, since $h^* \geq 1$. If it is nonfinite, the reserve is $d(q_U) + d(q_V) + \delta(h) - 2 \geq d(q_U) + d(q_V) - 1$. An active ground argument is not the bare functional F, so has reserve at

least four by induction. If the argument is inactive, the active operator cannot be bare F in this nonfinite case, whose first unit field is one. The reserve is again at least four.

For an active ground lambda with body U, the exact scalar is $vD_zq(U) + q(U)\{z := 1\}$, with the full-vector syntactic multiplier $v = 2\sum_i \mathrm{sz}^z([\![U]\!]_i) \geq 2$. There are $v+1$ active copies and v joining sum nodes. The transport result gives reserve at least $(v+1)d(q(U)) + v \geq 5$. It includes the positive-field size of a functional body. This completes the source induction and proves the claimed bounds. □

Lemma 6.4 (One final abstraction)**.** *For actual unary sources V_j free of a fresh x,*

$$[\![\lambda x.V_1(\cdots(V_m x)\cdots)]\!]_1 = \sum_{j=1}^{m}(V_{j,1}+3)+1.$$

Proof. The positive ground partial starts at one on x. One application adds the operator first field plus one from its whole-free positive partial; its scalar tail adds two more through its operator scalar and sum node. Thus it adds $V_{j,1}+3$. Induction on the list proves the formula, with one final lambda and no charge for reabstracting each copy. □

Theorem 6.5 (Original-budget ground-binder construction)**.** *Let $P : o$ be F-active in the complete grammar, $B = \lambda F.P$, and let $C : \sigma_k$ be an actual F-free caller. Write b for the second field of the original full box $[\![B]\!]\Box[\![C]\!]$. Then $b = B_2$ is finite, $b \geq 4$, and at $\mathcal{N} = 2^b$ one effectively constructs an actual unit-open unary K with*

$$\mathcal{N}(B_1 + C_1) = K_1 + \Delta, \qquad \Delta \in \mathbb{N}, \quad \Delta > 0.$$

No child construction certificate or enlargement of the original scale is assumed.

Proof. Since P is ground, B_1 is its positive scalar partial. Lemma 6.3 gives a multiset and $d \geq 4$ such that $B_1 = \sum_{V\in\mathcal{M}}(V_1+3)+d$. The output-two partial is finite by Lemma 6.1. At an active application that row is the head row plus the operator and argument scalar-partial rows and one, hence at least four. At an active lambda the same copying computation gives at least $2v+1 \geq 5$. The sole exception, bare F, is not ground. Thus $B_2 \geq 4$; the caller has level one, so the original untruncated second box is exactly B_2.

Unary-saturate the actual caller to $U_C = C0^{k-1}$, retaining $U_{C,1} = C_1$. In one flat list take $\mathcal{N}$ copies of every multiset occurrence and $\mathcal{N}$ copies of U_C. Lemma 6.4 gives

$$K_1 = \mathcal{N}\sum_{V\in\mathcal{M}}(V_1+3) + \mathcal{N}(C_1+3)+1, \quad \Delta = \mathcal{N}(d-3)-1 \geq \mathcal{N}-1 > 0.$$

Every carrier is actual by Lemma 6.2. Replication is finite and effective, though it may be enormous. There is no unsupported derivative realizer and no change to the original caller or its scalar value. □

7 From a certificate to the whole original head graph

The next result is a consumer with an explicit premise. It applies to the preceding theorem, but also to any other source construction paying the same premise.

Theorem 7.1 (Whole-head graph from a paid certificate)**.** *Suppose actual sources B, C, K and a finite original box b supply $a = 2^b(B_1 + C_1) = K_1 + \Delta$ at common functional units, where $\Delta \geq 0$ is finite and $K : o \to o$ is unit open. Then the complete genuine graph of*

$$J_\xi(t) = \psi(\xi + t - \mathrm{N}(\xi)), \quad \xi = \omega \otimes a, \quad t \geq \mathrm{N}(\xi),$$

is effectively existentially definable in actual-source arithmetic. It retains the original scalar/input wires of B, C and every finite-prefix input.

Proof. Close only auxiliary ground names of K to literal zero, obtaining K^0. Its positive field is unchanged by the source invariant; no ground name in the original BC is closed. After functional units set

$$k = K_1^0, \quad c = K_0^0, \quad \beta = \omega \otimes k, \quad C_\beta = \mathrm{N}(\beta) + c + 1.$$

These are fixed computable values. Theorem 5.2 observes the exact ray $[\![K^0 z]\!]_0 = J_\beta(C_\beta + n)$ for $z_0 = n + 1$, without replacing its real scalar c or inventing a unit inhabitant.

The certificate gives $\xi = \beta + \omega\Delta$ and $\mathrm{N}(\beta + \omega j) = \mathrm{N}(\beta) + 2j$. At a genuine total norm t, the exact predecessor of $\beta + \omega j$ is $\beta + \omega(j-1) + \Phi(t) - \mathrm{N}(\beta + \omega(j-1))$. The maximal-predecessor theorem of [4] proves this by retaining the affordable prefix and spending the rest on the finite tail. Consequently induction on j gives the *upward* identity

$$J_\xi(t) = J_\beta(\Phi^\Delta(t)) + \Delta \qquad (t \geq \mathrm{N}(\xi)).$$

This does not invert the sparse function Φ^Δ.

Set $T = \max(\mathrm{N}(\xi), C_\beta)$. For $t \geq T$ use the fixed finite chain of graph equations $u_0 = t$, $u_{j+1} = \Phi(u_j)$, and $u_\Delta = \mathrm{N}(\beta) + c + z_0$. Its required z_0 is positive because $u_\Delta \geq t \geq C_\beta$. Observe $v = [\![K^0 z]\!]_0$ through the unit-elimination theorem and put $y = v + \Delta$. The identity proves both existence and uniqueness of $y = J_\xi(t)$ on this tail.

Every Φ gate is definable by the actual S graph: $F_S(n) = \Phi(n + 4)$, with the four smaller inputs filled by computed constants. There is a fixed number Δ of gates, not an assumed variable-iteration primitive. For each genuine $t_0 < T$, use the exact closed evaluator to compute $j_0 = J_\xi(t_0)$ and include the disjunct $t = t_0 \wedge y = j_0$. These finitely many points and the tail define the entire genuine graph. Range constraints for the ray stay inside the tail branch; imposing them globally would lose small inputs.

The Boolean compiler handles this finite disjunction. Total observation graphs may be flattened outside it, but branch-specific range constraints may not. For the actual original target the shared raw wire is $t = \mathrm{N}(\xi) + B_0 + C_0 + \mathrm{lv}(\sigma)$; the unit-elimination theorem observes these scalars without closing their original ground names. Inserting this numerical wire changes no original source or valuation. All closed evaluations, replication, literal lengths and finite-prefix enumeration are charged to effectivity, with no feasible complexity assertion. □

The grammar restriction is real. With $\tau = o \to o$ and $H = R^o 0(D^o 0) : \tau$, one has $H_1 = 2\omega$. For $F : \tau \to \tau$, the first field of FH is $h = 2^{F_2} \otimes (F_1 + 2\omega)$. Its unit value is $4\omega + 2$, while its positive F partial is $2\omega + 6$. No finite reserve makes these equal. Thus an active functional argument can defeat the affine invariant; the example does not refute every possible source-construction method.

8 Functional inputs at the unchanged control

The standard control can already support an undecidable comparison problem when free functional variables are admitted. This is a different domain from the all-ground question. We give the reduction because the distinction identifies exactly where its additional arithmetic information enters.

The published undecidable problem used here is solvability of a finite system of linear equations

$$\sum_i a_{ji}(X)p_i(X) - g_j(X) = 0, \qquad a_{ji}, g_j \in \mathbb{Z}[X], \quad p_i \in \mathbb{N}[X].$$

It is linear in the unknown polynomials, not a system of arbitrary Diophantine products of unknowns. Narendran's theorem and Dong's discussion give this precise polynomial-semiring boundary [2, 3]. Zero polynomials are allowed. We use undecidability, without importing a stronger many-one completeness classification of that source problem.

Put $\tau = o \to o$. For actual sources $A, B : \tau$ define

$$\begin{aligned} \mathsf{P}(A,B) &= (\lambda z : \tau.A)B, \qquad \mathsf{H}(A) = R^o 0(\lambda n : o.A), \\ \mathsf{S}(A,B) &= (\lambda z : (o \to \tau).A)(\lambda n : o.B). \end{aligned}$$

The original argument order is $R^\rho : o \to (o \to \rho \to \rho) \to \rho \to \rho$: its first argument $0 : o$ is the recursion test, not its base value. Thus $R^\rho 0(\lambda n : o.A)$ has type $\rho \to \rho$ when $A : \rho \to \rho$, also at the higher type used below. The displayed bound names are unused and fresh. Writing $a_i = [\![A]\!]_i$, $b_i = [\![B]\!]_i$, the literal canonical clauses give

$$\begin{aligned} [\![\mathsf{P}(A,B)]\!]_1 &= 8a_1 + 4b_1 + 8, \\ [\![\mathsf{S}(A,B)]\!]_1 &= 8a_1 + 8b_1 + 16, \\ [\![\mathsf{S}(A,B)]\!]_0 &= \psi\big(\omega \otimes (8a_1 + 8b_1 + 16) + 9a_0 + 5b_0 + 13\big), \\ [\![\mathsf{H}(A)]\!]_1 &= 2\omega \otimes a_1 + 3\omega. \end{aligned}$$

For example the unused τ binder has vector $(9a_0 + 8, 2a_1 + 2, 2)$, and its box with B gives the first formula. For the second binder its scalar is again $9a_0 + 8$ and its first field $2a_1 + 2$; the ground lambda on B has vector $(5b_0 + 4, 2b_1 + 2)$, giving both S formulas. The recursor prime followed by its two displayed arguments gives the H formula. These calculations use actual source syntax; they do not introduce an arbitrary ordinal-valued primitive.

Lemma 8.1 (Signed polynomial pairs)**.** *Every signed linear expression D in ordinal unknowns with coefficients in $\mathbb{Z}[\omega]$ has effectively constructed actual $A_D, B_D : \tau$ whose first-coordinate difference is $2^k D$ for a finite k. All free functional variables have type τ, and internal types have level at most two. Differences are taken only in the signed Cantor-monomial group; each of the two source coordinates remains a positive ordinal.*

Proof. Let $C = \lambda u : o.u$, with $C_1 = 1$. The first fields of $\mathsf{P}(C,C)$ and $\mathsf{S}(C,C)$ are 20 and 32. Take sixteen pairs: eight copies of $(\mathsf{S}(C,C), \mathsf{P}(C,C))$ of difference 12, five of $(C, \mathsf{P}(C,C))$ of difference -19, and three of (C,C). Combine them componentwise in a balanced S tree of height four. Constants cancel in every paired combination, so the resulting difference is

$$8^4(8 \cdot 12 - 5 \cdot 19) = 2^{12}.$$

For a free $X : \tau$, the pair $(\mathsf{P}(X,C), \mathsf{P}(C,C))$ has difference $8(x_1 - 1)$. Three multiplications by eight followed by a paired S combination with the constant pair give $2^{15}x_1$.

The pair operation $(A,B) \mapsto (\mathsf{S}(A,C), \mathsf{S}(B,C))$ multiplies a difference by eight, and $(A,B) \mapsto (\mathsf{S}(A,A), \mathsf{S}(B,B))$ multiplies it by sixteen. Applying H to both entries multiplies a difference by 2ω; swapping entries negates it. Repeated addition constructs every finite integer coefficient. To add pairs with scale exponents k_1, k_2, choose $K \geq \max(k_1, k_2) + 6$. Every integer at least six is $3u + 4v$ with $u, v \geq 0$, so the two scale operations align both differences at 2^K. A paired S then gives their sum with scale 2^{K+3}. Induction on the finitely many signed monomials proves the assertion. The displayed source combinators introduce no internal level above two. □

The two scalar coordinates of an equation pair need not agree when their positive coordinates agree. We therefore pay that alignment explicitly.

Lemma 8.2 (Alignment and a full-vector detector)**.** *The condition* $[\![A]\!]_1 = [\![B]\!]_1$ *is equivalent to existence of two fresh positive ground inputs making the full canonical vectors of*

$$A' = (\lambda t : o.A)z, \qquad B' = (\lambda t : o.B)w$$

equal. Full-vector equality of any actual pair of type τ *can in turn be detected by ground sources* L, R *satisfying* $L_0 \geq R_0$*, with equality exactly at vector equality and otherwise gap at least two.*

Proof. The aligned first fields are $2a_1 + 2$ and $2b_1 + 2$, and their scalars are

$$\psi(\omega \otimes (2a_1 + 2) + 5a_0 + 4 + z_0), \qquad \psi(\omega \otimes (2b_1 + 2) + 5b_0 + 4 + w_0).$$

Equal vectors imply $a_1 = b_1$. If this equality holds, choose $z_0 = 5b_0 + 1$, $w_0 = 5a_0 + 1$. Both tails and both vectors then agree. These choices are made after the original valuation, so are not circular.

For the detector, now write A, B for the pair being tested, and form

$$X = \mathsf{S}(A, A), \quad Y = \mathsf{S}(B, B), \quad Z = \mathsf{S}(A, B), \quad W = \mathsf{S}(B, A).$$

Make the actual shared ground payload $Q = W(Z(Y(X0)))$, and put $q = Q_0$. The unconditional finite-tail lower bound for ψ gives

$$q \geq 1 + x_0 + y_0 + z_0 + w_0 \geq 5.$$

Set $g_X = XQ$, and similarly g_Y, g_Z, g_W, and finally set $L = e(g_X, g_Y)$, $R = e(g_Z, g_W)$.

Suppose first $a_1 > b_1$. Then $x_1 > z_1 = w_1$. The ordinals

$$\alpha = \omega \otimes z_1 + z_0 + q + 2, \qquad \beta = \omega \otimes x_1 + x_0 + q$$

satisfy $\alpha < \beta$. We must check a norm premise before comparing their collapses. Bounded norms of actual assignments and the published natural-product norm inequality [1, Propositions 2.2–2.3] give

$$\mathrm{N}(\alpha) \leq 2\,\mathrm{N}(z_1) + z_0 + q + 2 \leq 3z_0 + q + 2 \leq 4q + 2 \leq 2^{q+1} \leq \Phi(\mathrm{N}(\beta)).$$

Controlled strictness therefore gives $\psi(\beta) > \psi(\alpha)$. Two consecutive fixed-limit tail-growth steps from [4] yield $g_{X,0} > 4g_{Z,0}$. The same calculation with W gives $g_{X,0} > 4g_{W,0}$. If $b_1 > a_1$, use Y in place of X.

If $a_1 = b_1$ but $a_0 \neq b_0$, the four S heads coincide. Their scalar tails are

$$14a_0 + 13, \quad 14b_0 + 13, \quad 9a_0 + 5b_0 + 13, \quad 9b_0 + 5a_0 + 13.$$

The larger diagonal tail exceeds both cross tails by at least five. Strict increase gives at least a two-unit scalar gap, so two fixed-tail growth steps in the subsequent applications give again a diagonal g value greater than four times each cross value.

In either unequal case, $3g_{X,0} + g_{Y,0} > 3g_{Z,0} + g_{W,0}$: even when the larger diagonal is the one with coefficient one, its fourfold domination suffices. The final eraser is increasing and has gaps at least two. Equal input vectors give four equal S vectors, hence equality. This proves the claimed detector on every admissible valuation, not only on the specially chosen alignment witnesses. □

Theorem 8.3 (Standard-control functional-input undecidability)**.** *Full universal canonical weak comparison and strict comparison are undecidable already for ground output, free-variable levels at most one, and internal levels at most two. The reduction at this level uses finitely many comparison queries. A computable single-query reduction from the same polynomial-system unsatisfiability problem is available with internal level at most three.*

Proof. Let π retain exactly the Cantor monomials with finite exponents. It maps ordinals below ε_0 to $\mathbb{N}[\omega]$. It preserves natural sum and multiplication: a natural sum of exponents is finite exactly when both exponents are finite. It consequently extends to the signed Cantor-monomial group and respects multiplication by fixed $\mathbb{Z}[\omega]$ coefficients.

For the level-two assertion, enumerate the finitely many possible sets of zero polynomial unknowns, substitute zero for those, and require the others to be positive ordinal first coordinates. Compile the resulting equations by Lemma 8.1, align them separately, and apply Lemma 8.2. Aggregate the nonnegative ground detectors in identically shaped eraser trees. Root equality is exactly simultaneous equation satisfaction. Its absence is one universal strict comparison; the $KR \preccurlyeq IL$ conversion in Theorem 3.1 gives the same weak query. An empty system receives a tautology.

A polynomial solution lifts in the matching zero pattern: each remaining nonzero polynomial is an admissible positive first field, after choosing its zeroth field sufficiently large for its norm. Conversely any positive-ordinal solution projects by π to a polynomial solution of the original system. Some projected unknowns may become zero, which is permitted in that system. Thus an algorithm for either comparison would decide polynomial solvability using the finitely many patterns, contradicting the published theorem.

For a single query choose instead a fixed positive ordinal Ω with $\pi(\Omega) = 0$, and transform each original equation to

$$\sum_i a_i(\omega)x_i - g(\omega) - \left(\sum_i a_i(\omega)\right)\Omega = 0.$$

A polynomial solution u_i lifts by $x_i = u_i \oplus \Omega > 0$. Conversely projection of any positive-ordinal solution solves the original equations. The transformation preserves simultaneous use of each unknown.

We must realize the extra constant in actual source syntax. Put $\sigma = \tau \to \tau$, $C_2 = \lambda u : \tau.u$, and $\mathsf{H}_2(A) = R^\tau 0(\lambda n : o.A)$. The canonical clauses give $(C_2)_2 = 2$ and $(\mathsf{H}_2(A))_2 = \omega + 2\omega \otimes A_2$. The closed $H = \mathsf{H}_2(\mathsf{H}_2(C_2))$ has

$$H_2 = 10\omega^2 + \omega, \qquad \Omega = 2^{10\omega^2+\omega} = \omega^{10\omega+1}, \qquad \pi(\Omega) = 0.$$

Apply this same actual H to the two entries of the constant pair of Lemma 8.1. The box gives $[\![HA]\!]_1 = \Omega \otimes (H_1 + A_1)$, so their difference is $2^{12}\Omega$. Add this as a second closed seed to the pair compiler. Only fixed ω shifts and signed sums of that seed are needed. Compile the one transformed system and apply the same alignment and detector. The closed R^τ raises the internal level bound to three, while every free variable still has level at most one. This is the promised single-query reduction. No claim about all-ground free variables follows. □

9 An additive-one change of control

Growth alone does not explain the positive standard-control results in [5]. Their modular hypotheses carry indispensable information. The following construction changes the control while retaining its growth, any prescribed finite prefix, and even pointwise additive accuracy.

Fix a primitive recursive bounded simulation predicate $H(e,t)$ for an effective enumeration of blank-input programs: it is one precisely when program e has halted within t steps. For a fixed finite b, define

$$\epsilon_b(0) = 0, \qquad \epsilon_b(n) = \begin{cases} 0 & 0 < n \le b, \\ H(v_2(n), n) & n > b, \end{cases} \qquad \Gamma_b(n) = \Phi(n) + \epsilon_b(n),$$

where $v_2(n)$ is the exponent of two dividing $n > 0$. The one function Γ_b is fixed before a program index is supplied.

Lemma 9.1 (Control properties)**.** *The function Γ_b is primitive recursive, agrees with Φ on $[0,b]$, and satisfies*

$$0 \le \Gamma_b(n) - \Phi(n) \le 1, \quad \Phi(n) \le \Gamma_b(n) < \Phi(n+1), \quad \Gamma_b(n+1) > 4\Gamma_b(n).$$

It has the additive margin $\Gamma_b(v+a) \ge \Gamma_b(v) + 2a$ for $a \ge 1$, is eventually bounded by F_6, and $\Gamma_b(n)/\Phi(n) \to 1$.

Proof. Bounded simulation, valuation v_2, and each fixed F_j are primitive recursive. The prefix and additive estimates follow from the definition. If $u = \Phi(n)$, the control estimate in [4] gives $\Gamma_b(n+1) \ge 2^{2^u} > 4(u+1) \ge 4\Gamma_b(n)$. It also gives $\Phi(n) + 1 < \Phi(n+1)$ and increments greater than two. Sum those increments for the additive margin. For $n \ge 101$, $F_5(n+101) \le F_5(F_5(n)) \le F_6(n)$, since $F_5(n) \ge n + 101$; this bounds $\Gamma_b(n) < F_5(n+101)$. The bounded error and $\Phi(n) \to \infty$ give the ratio limit. □

These estimates supply the identity-majorization and additive margin of the collapse construction, as well as its eventual finite-hierarchy upper bound. Only the coefficient-one collapse will be used for hardness. For a strictly increasing identity-majorizing control its exact formula is $\psi_{\Gamma_b}(\omega+n) = \Gamma_b(n+2)+1$, by the maximum-predecessor recurrence. It is injective. Its consecutive values exceed four times the preceding ones at the arguments used by the eraser. Consequently the source addition, constants, collision detector and conjunction construction proved above work verbatim with this control. We do not transfer a standard-control residue theorem to Γ_b.

Theorem 9.2 (Ground hardness within distance one)**.** *For every b, universal canonical weak comparison and strict comparison at the one fixed control Γ_b are Π^0_1-complete already on ground-output sources having only ground free names and ground binders, using only the prime constants $0, S$, with internal levels at most one. Moreover $\Gamma_b(n) \bmod 2$ is not eventually periodic.*

Proof. Every value of F_j at a positive argument, and thus every value of Φ, is a power of two divisible by two. This follows by induction from its base and its positive number of iterations. For $n > b$ the value $\Gamma_b(n)$ is therefore odd exactly when $H(v_2(n), n) = 1$.

Fix the odd constant $C = 2b+5$. For a program index e consider the existential conjunction over natural variables

$$z + 4 = 2^e(2t + C), \qquad y = F_S(z), \qquad y = w + w + 1.$$

Here $F_S(z) = \Gamma_b(z+4)$, and 2^e is a fixed coefficient computed from the input index. It is not a variable exponentiation operation. With $n = 2^e(2t+C)$ one has $n \ge 5$, $n > b$ and $v_2(n) = e$. The conjunction has a solution at t exactly when $H(e,n) = 1$. If e halts, choose t large enough that n exceeds its halting time; then $z = n - 4 \ge 0$ and $w = (\Gamma_b(n) - 1)/2 \ge 0$ supply

witnesses. Conversely any solution certifies the bounded halting test. Thus the conjunction is satisfiable exactly when program e halts.

Flatten it using the proved addition, constant and actual S graph gadgets, then use the collision detector and eraser aggregation. Its unsatisfiability is one universal strict comparison and, by $KR \preccurlyeq IL$, one weak comparison. All introduced binders are ground, and all leaves use only $0, S$ and ground variables. The nonhalting reduction proves Π_1^0 hardness with the stated source restrictions. The arithmetic DAG uses $O(e + \log(b + 2))$ gates; this is a bound in the numerical index e, not its binary length.

For membership, enumerate all finite tuples of positive raw ground values. Every value of a displayed source is computable: its closed heads are finite and the collapse reduces to fixed finite iteration of the primitive recursive control. A failed universal comparison is therefore semidecidable. This proves completeness, without an oracle hidden in Γ_b.

Finally suppose parity were periodic with period $p \geq 1$ after N. For each e effectively choose $n_e > \max(N, b)$ with $v_2(n_e) = e$. Every point

$$n_e + kp2^{e+1}, \qquad k = 0, 1, 2, \ldots,$$

has the same residue modulo p and the same valuation e. Periodicity would make the bounded tests $H(e, n_e + kp2^{e+1})$ constant. They are always zero if e never halts and eventually one if it halts. Computing $\Gamma_b(n_e) \bmod 2$ would thus decide halting. The alleged finite N, p can be hardwired; they need not be found uniformly from b for this contradiction. No eventual parity period exists. □

The functional-input theorem leaves the control unchanged and changes the input domain. The last theorem keeps ground inputs but changes the control. Neither is a negative answer for the remaining exact-standard-control, all-ground canonical comparison problem. Together with the positive standard-control classes, they explain why a claimed solution must preserve both the source domain and the precise arithmetic function.

10 The remaining exact-control boundary

The all-ground standard-control comparison problem remains open. The functional-input theorem changes the permitted free-variable vectors; the control-sensitivity theorem changes the numerical control. The source realization theorem closes a complete constructor grammar, but not arbitrary active functional argument application. Their distinct hypotheses are part of the results, not removable technical decorations.

For a supplied actual unary source, the complete-count residue-attainment reduction in [4] uses Theorem 3.1 and proves a necessary arithmetic subproblem. It does not prove that every auxiliary ordinal head or every isolated intermediate partition coefficient has the same interpretation. A full solution must either decide the existential actual-source systems themselves or interpret a known undecidable problem there while preserving the unchanged source and control contracts.

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